بسم الله الرحمن الرحيم

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS

1.1 NUMBERS UP TO ONE BILLION

Read numbers up to one billion in numerals and in words

Concept of One Billion

We have learnt in class IV that the smallest number of nine digits is **100000000** (One hundred million)

Look at the number "34,768,172"

Millions	Thousands	Ones
34	768	172

- How many millions are in **34,768,172**?
- How many thousands are in **34,768,172**?
- How many ones are in **34,768,172**?
- Write, **34,768,172** in words.

____

____

The greatest number of nine digits is 999,999,999 read as "Nine hundred ninety nine million, nine hundred ninety nine thousand and nine hundred ninety nine". When we add one more to it, then

999,999,999 + 1 = 1,000,000,000

Teacher's Note After revision of numbers in millions, teacher should help students to develop the concept of one billion by counting numbers of digits by using flash cards or other related material.

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS
(Numbers up to one billion)

The number after **999,999,999** is **1,000,000,000**. It is read as One thousand million or 1 billion..

One Billion = One thousand million

According to the place value chart of numbers, there are the following four periods for representing numbers upto billions.

4th period	3rd period	2nd period	1st period
Billions ,	Millions ,	Thousands ,	Ones

Representation of one billion in Place Value Chart

Billions	Millions			Thousands			Ones		
B	H-M	T-M	M	H-Th	T-Th	Th	H	T	O
1	0	0	0	0	0	0	0	0	0

Here **H-M** means Hundred Million.
To read a number, we put commas to after every three digits from the right to separate the periods

Example. Read the following numbers:

(i) **245612384** (ii) **1000000000**

Solution: **(i) 245612384**

We first separate the digits of numbers in periods.

245, 612, 384

Millions	Thousands	Ones
245	612	384

We read it as "Two hundred forty five million, six hundred twelve thousand and three hundred eighty four" and write it as: 245,612,384.

Solution: (ii) 1000000000

We first separate the digits of numbers in periods:

Billions	Millions			Thousands			Ones		
1	0	0	0	0	0	0	0	0	0

We can read it as: "One Billion"

1.1.2 Write numbers up to 1,000,000,000 (one billion) in numerals and in words.

Example 1. **Write 234,567,812 in words.**

Solution. We first write the digit of the numbers in place value chart.

234,567,812

Millions			Thousands			Ones		
H-M	T-M	M	H-th	T-th	Th	H	T	O
2	3	4	5	6	7	8	1	2

Hence the given number is 234,567,812 and written in words as Two hundred thirty four million, five hundred sixty seven thousand and eight hundred twelve.

Example 2. **Write "Three hundred fifty six million, two hundred sixty seven thousand and nine hundred two" in numerals.**

Solution: We first write the number in the place value chart:

Millions			Thousands			Ones		
H-M	T-M	M	H-Th	T-Th	Th	H	T	O
3	5	6	2	6	7	9	0	2

So, the required number is **356,267,902**.

EXERCISE 1.1

A. Write the digit under their correct place value and also write the number in words.

3 thousand
6 hundred
7 ten million
8 ten thousand
1 million
0 ones
1 hundred million
8 tens
2 hundred thousand

B. Separate by periods and read the following numbers.

(1) 45672 (2) 2670273 (3) 34296127
(4) 100000000

C. Write the following numbers in words.

(1) 66,655,522 (2) 96,340,529 (3) 245,672,316
(4) 100,000,000

D. Write the following numbers in numerals.

(1) One million, two thousand and six hundred.

(2) Nine million, ninety nine thousand and seventy seven.

(3) Fifty eight million, eight hundred sixty two thousand and forty five.

(4) One billion

(5) Three hundred forty five million, six hundred seventy one thousand and eight hundred six.

1.2 ADDITION AND SUBTRACTION

1.2.1 Add numbers of complexity and of arbitrary size

We have learnt to add and subtract numbers up to 6 digits.
Let us revise.

We always start addition or subtraction from ones.

Example 1.

Add 638,941 and 347,036

Write numbers according to place values and then add.

Solution:

	H-Th	T-Th	Th	H	T	O
		1				
	6	3	8	9	4	1
+	3	4	7	0	3	6
	9	**8**	**5**	**9**	**7**	**7**

Example 2.

Add 359,990 and 406,780

Write numbers according to place values and then add.

Solution:

	H-Th	T-Th	Th	H	T	O
		1	1	1		
	3	5	9	9	9	0
+	4	0	6	7	8	0
	7	**6**	**6**	**7**	**7**	**0**

While adding numbers we should write the digits according to their place value and then add them accordingly.

Example 3.

Add: 5,389,624 and 930,157

Solution:

	M	H-Th	T-Th	Th	H	T	O
	1	1				1	
	5	3	8	9	6	2	4
+		9	3	0	1	5	7
	6	**3**	**1**	**9**	**7**	**8**	**1**

Example 4.

Fill the missing digits of the following.

	6	4	5	1	◯	3
+	3	◯	3	◯	1	2
	9	**8**	**8**	**6**	**3**	**5**

EXERCISE 1.2

A. Solve.

(1) 713492 + 268310

(2) 4318114 + 313934

(3) 8193860 + 429177

(4) 6216921 + 834538

(5) 3941234 + 590543

(6) 5491869 + 605304

(7) 7390443 + 1538006

(8) 2555312 + 1084881

(9) 5319432 + 4181518

(10) 34653635 + 2660477

(11) 56018987 + 23594431

(12) 438394568 + 351056939

B. Add.

(1) 680,563 and 563,168

(2) 3,541,371 and 232,164

(3) 34,399,188 and 6,412,508

(4) 65,943,022 and 22,913,924

(5) 840,233,419 and 65,113,846

(6) 733,050,195 and 188,439,919

C. Fill the missing digits in the following.

(1)

	5	3	6	2	◯	4
+	3	◯	3	◯	2	2
	8	8	9	6	3	6

(2)

	3	4	7	8	4	5
+	4	◯	2	◯	1	◯
	7	6	0	1	5	9

(3)

	5	6	0	1	8	9	7
+	2	◯	5	◯	4	◯	1
	8	1	5	5	3	1	8

(4)

	3	4	◯	5	◯	3	7
+	2	◯	4	2	0	◯	2
	6	0	6	7	1	4	9

D. Faisal's father was admitted to the hospital for 1 day for an operation. His operation charges were Rs. 350,000, the room charges were Rs. 1200 and his medicine charges were Rs. 125,500. How much was the total bill?

1.2.2 Subtract numbers of complexity and of arbitrary size

Example 1.

Subtract 430,912 from 871,032

Write numbers according to place values and then subtract.

Solution:

	H-Th	T-Th	Th	H	T	O
			0	10		
	8	7	~~1~~	0	3	2
–	4	3	0	9	1	2
	4	4	0	1	2	0

Example 2.

Subtract 273,587 from 307,843.

Write numbers according to place values and then subtract.

Solution:

	H-Th	T-Th	Th	H	T	O
		10		7	13	13
	3	~~0~~	7	~~8~~	~~4~~	~~3~~
–	2	7	3	5	8	7
	0	3	4	2	5	6

Numbers of more than 6-digits can be subtracted in the same way.

Example 3. Subtract 6,134,248 from 8,206,884

Solution:

	M	H-Th	T-Th	Th	H	T	O
		1	10			7	10
	8	~~2~~	0	6	8	~~8~~	4
–	6	1	3	4	2	4	8
	2	**0**	**7**	**2**	**6**	**3**	**6**

Explanation:

(i) Borrow 1-ten from 8 tens, add 4 ones to get 14 ones; where 7 tens are left. Now complete the process of subtraction up to thousands.

(ii) Again borrow 1 hundred-thousand from 2 hundred-thousands and add to ten-thousands to get 10 ten-thousands, where 1 hundred thousand is left. In this way complete the process of subtraction.

Activity 1

Fill the missing digits in the following.

	5	8	5	4	6	3
–	3	◯	1	◯	1	2
	2	**3**	**4**	**2**	**5**	**1**

Activity 2

Fill the missing digits in the following.

	7	4	◯	8	6	◯
–	3	◯	1	◯	1	2
	4	**4**	**2**	**4**	**5**	**3**

EXERCISE 1.3

A. Solve:

(1) $\begin{array}{r} 2863132 \\ -\ 164350 \\ \hline \end{array}$ (2) $\begin{array}{r} 5634153 \\ -\ 393844 \\ \hline \end{array}$ (3) $\begin{array}{r} 4194312 \\ -\ 994208 \\ \hline \end{array}$

(4) $\begin{array}{r} 4395684 \\ -\ 2348736 \\ \hline \end{array}$ (5) $\begin{array}{r} 50862131 \\ -\ 944029 \\ \hline \end{array}$ (6) $\begin{array}{r} 65309949 \\ -\ 8214309 \\ \hline \end{array}$

B. Subtract.

(1) 214,379 from 600,500 (2) 856,394 from 3,767,595

(3) 4,930,109 from 5,851,036 (4) 5,396,138 from 43,547,967

(5) 35,180,962 from 89,086,371 (6) 134,258,369 from 656,148,154

C. Fill the missing digits in the following.

(1)
$$\begin{array}{r} 8\;7\;3\;2\;4\;6 \\ -\;4\;3\;\bigcirc\;\bigcirc\;2\;\bigcirc \\ \hline \boxed{4\;4\;2\;1\;2\;3} \end{array}$$

(2)
$$\begin{array}{r} 6\;9\;4\;7\;\bigcirc\;2 \\ -\;3\;\bigcirc\;3\;\bigcirc\;2\;4 \\ \hline \boxed{3\;8\;1\;6\;3\;8} \end{array}$$

(3)
$$\begin{array}{r} 5\;\bigcirc\;3\;4\;\bigcirc\;9\;8 \\ -\;3\;2\;\bigcirc\;2\;7\;4\;\bigcirc \\ \hline \boxed{2\;7\;2\;1\;5\;5\;4} \end{array}$$

(4)
$$\begin{array}{r} 6\;8\;1\;4\;\bigcirc\;3\;9 \\ -\;\bigcirc\;\bigcirc\;0\;2\;5\;2\;\bigcirc \\ \hline \boxed{3\;4\;1\;2\;2\;1\;5} \end{array}$$

D. Pakistan imported 107,857 kg dry grapes in 2014, and 154,792 kg dry grapes in 2015. How many more dry grapes were imported in 2015 than in 2014?

1.3 MULTIPLICATION AND DIVISION

1.3.1 Multiply numbers, up to 6-digits, by 10,100 and 1000

We have learnt in class IV to multiply numbers. Let us revise. Remember that multiplication is the process of continuous addition of the same number many times.

Example 1. Multiply: (i) 2,658 by 10 (ii) 38,524 by 10

(iii) 451,392 by 10

Solution:

$$\begin{array}{rrrrr} & \text{Th} & \text{H} & \text{T} & \text{O} \\ & 2 & 6 & 5 & 8 \\ & & \times & 1 & 0 \\ \hline & 0 & 0 & 0 & 0 \\ +\,2 & 6 & 5 & 8 & \times \\ \hline \boxed{2} & \boxed{6} & \boxed{5} & \boxed{8} & \boxed{0} \end{array}$$

$$\begin{array}{rrrrrr} & \text{T-Th} & \text{Th} & \text{H} & \text{T} & \text{O} \\ & 3 & 8 & 5 & 2 & 4 \\ & & & \times & 1 & 0 \\ \hline & 0 & 0 & 0 & 0 & 0 \\ +\,3 & 8 & 5 & 2 & 4 & \times \\ \hline 3 & 8 & 5 & 2 & 4 & 0 \end{array}$$

$$\begin{array}{rrrrrrr} & \text{H-Th} & \text{T-Th} & \text{Th} & \text{H} & \text{T} & \text{O} \\ & 4 & 5 & 1 & 3 & 9 & 2 \\ & & & & \times & 1 & 0 \\ \hline & 0 & 0 & 0 & 0 & 0 & 0 \\ +\,4 & 5 & 1 & 3 & 9 & 2 & \times \\ \hline 4 & 5 & 1 & 3 & 9 & 2 & 0 \end{array}$$

From the examples, it is clear that in the multiplication of a natural number with 10, we put a zero in the given number at right and rest of the number remains same. This rule can also be applied in the multiplication of numbers with 100 and 1000.

Example 2. Multiply

(i) **4,353 by 100**

(ii) **58,351 by 1,000**

Solution: (i)

			Th	H	T	O
			4	3	5	3
			x	1	0	0
			0	0	0	0
		0	0	0	0	x
+	4	3	5	3	x	x
	4	**3**	**5**	**3**	**0**	**0**

Solution: (ii)

				T-Th	Th	H	T	O
				5	8	3	5	1
				x	1	0	0	0
				0	0	0	0	0
			0	0	0	0	0	x
		0	0	0	0	0	x	x
+	5	8	3	5	1	x	x	x
	5	**8**	**3**	**5**	**1**	**0**	**0**	**0**

Example 3. Multiply 536,847 by 10, 100 and 1,000.

Solution:

H-Th	T-Th	Th	H	T	O			HM	TM	M	H-Th	T-Th	Th	H	T	O
5	3	6	8	4	7	x 10	=			**5**	**3**	**6**	**8**	**4**	**7**	**0**
5	3	6	8	4	7	x 100	=		**5**	**3**	**6**	**8**	**4**	**7**	**0**	**0**
5	3	6	8	4	7	x 1000	=	**5**	**3**	**6**	**8**	**4**	**7**	**0**	**0**	**0**

Remember: To multiply a number by 10 (100), (1000); we write one zero, (two zeroes), (three zeroes) at the right of the given number. The rest of the digits remain the same.

Teacher's Note Teacher should ensure enough practice of multiplication by 10, 100 and 1000 by asking questions orally.

1.3.2 Multiply numbers, up to 6-digits, by a 2-digits and 3-digits numbers

Example 1. **Multiply: (i) 754,863 by 40 (ii) 754,863 by 400**

Solution:

(i)

7 5 4 8 6 3 x 40
= 7 5 4 8 6 3 x **4** x **10**
= (754863 x **4**) x **10**
= 3019452 x **10** = 30194520
= 30,194,520

In case of multiplication of a number with 40, we have to multiply it with 4 and 10. So, multiply the given number by 4 and put one zero in the right of the product for multiplying by 10.

(ii)

7 5 4 8 6 3 x 400
= 7 5 4 8 6 3 x **4** x **100**
= (754863 x **4**) x **100**
= 3019452 x **100** = 301945200
= 301,945,200

Similarly in case of multiplication of number with 400, we have multiplied the given number by 4 and put two zeroes in the right of the product for multiplying by 100.

Example 2. **Multiply 323,114 by 32**

Solution:

```
              1
    3  2  3  1  1  4
             x  3  2
--------------------
    1
    6  4  6  2  2  8   → Multiplication by 2 ones (1)
  1
+ 9  6  9  3  4  2  x  → Multiplication by 3 tens (10)
--------------------
1 0 , 3  3  9 , 6  4  8
```

Thus, 323114 x 32 = 10,339,648

Example 3. **Find the product of 230214 and 103**

Solution:

$$
\begin{array}{rcccccccc}
 & & & & & & 1 & & \\
 & & & 2 & 3 & 0 & 2 & 1 & 4 \\
 & & & & \times & 1 & 0 & 3 & \\
\hline
 & & & 1 & & 1 & & & \\
 & & & 6 & 9 & 0 & 6 & 4 & 2 \\
 & & 0 & 0 & 0 & 0 & 0 & 0 & \times \\
+ & 2 & 3 & 0 & 2 & 1 & 4 & \times & \times \\
\hline
 & \boxed{2} & \boxed{3} & \boxed{7} & \boxed{1} & \boxed{2} & \boxed{0} & \boxed{4} & \boxed{2}
\end{array}
$$

Thus, 230214 x 103 = 23,712,042

EXERCISE 1.4

A. Solve.

(1) 4,136 x 10
(2) 34,569 x 10
(3) 21,034 x 10
(4) 15,347 x 100
(5) 27,796 x 100
(6) 155,430 x 100
(7) 41,357 x 1000
(8) 386,975 x 1000

B. Solve.

(1) 1,942 x 50
(2) 63,578 x 80
(3) 25,608 x 70
(4) 326,985 x 90
(5) 8,540 x 300
(6) 280,915 x 600

C. Multiply.

(1) 25,839 x 33
(2) 243,419 x 86
(3) 65,204 x 75
(4) 467808 x 92
(5) 76,391 x 22
(6) 298,543 x 44
(7) 349,776 x 53
(8) 531,062 x 68
(9) 12,873 x 425
(10) 859,046 x 710
(11) 357,904 x 486
(12) 809,507 x 572
(13) 598,722 x 235
(14) 914,076 x 572
(15) 743,158 x 377
(16) 865, 432 x 444

D. The school library ordered 45 books. The cost of each book was Rs. 325. What is the cost of 45 library books?

1.3.3 Divide numbers, up to 6-digit s, by a 2-digit and 3-digit numbers

Let us consider the following examples.

Example 1. Divide 643185 by 24 and find quotient and remainder.

```
            26799     ⟶ Quotient
Divisor ⟵ 24) 643185  ⟶ Dividend
            - 48
            ----
              163
            - 144
            -----
              0191
            -  168
            ------
               0238
            -   216
            -------
               0225
            -   216
            -------
Remainder ⟵     009
```

Remainder should be less than the divisor

Step 1: Since digit of highest place value 6 is less then divisor 24. So, we take 64 multiply 24 by 2. Write 2 as quotient and subtract (64 – 48 = 16)

Step 2: 16 is the remainder so after bringing down 3 we get 163. Multiply 24 by 6 and get 144. Write 6 as quotient and subtract (163 – 144 = 19)

Step 3: The next digit in dividend is 1. So, we place it after 19, we get 191. Multiply 24 by 7. Write 7 as quotient and subtract (191 – 168 = 23).

Step 4: Next digit in dividend is 8, we place it after remainder 23. We get 238. Multiply 24 by 9, we get 216. Write 9 as quotient and subtract (238 – 216 = 22).

Step 5: Next digit in dividend is 5, so we put it after remainder 22 and get 225. Multiply 24 by 9, we get 216. Write 9 as quotient and subtract (225 – 216 = 9).

Hence quotient is **26,799** and remainder is **9**.

Example 2. Divide 837576 by 123

Solution:

```
         6809
  123) 837576
      - 738
      -----
        0995
      -  984
      ------
       01176
      - 1107
      ------
          69
```

Hence quotient is **6809** and remainder is **69**.

EXERCISE 1.5

A. Divide:

(1) 295,845 by 33
(2) 569,551 by 89
(3) 639,133 by 97
(4) 576,480 by 60
(5) 269,760 by 480
(6) 135,095 by 205
(7) 444,771 by 321
(8) 466,896 by 822

B. Find the remainder and quotient of the following.

(1) 5,678 ÷ 10
(2) 396,785 ÷ 10
(3) 473,405 ÷ 100
(4) 843,216 ÷ 100
(5) 5,230,106 ÷ 100
(6) 8,256,879 ÷ 1000
(7) 6,456,782 ÷ 1000
(8) 9,650,000 ÷ 1000

C. Mr. Anwar cooks 61500 *naans* in 500 days. How many *naans* must he cook every day if he cooks equal number of *naans* daily?

1.3.4 Solve real life problems involving mixed operations of addition, subtraction, multiplication and division

Example 1. Najeeb spends Rs 438900 on buying a house and Rs 358400 on buying a car. How much did he spend in all?

Solution: This problem involves addition.

Amount spent on house		4	^{1}3	^{1}8,	9	0	0
Amount spent on car	+	3	5	8,	4	0	0
Total amount		**7**	**9**	**7,**	**3**	**0**	**0**

Hence Najeeb spent **Rs797,300**

Example 2. According to census report 1998, there were 2,380,463 females and 1,511,021 males in Hyderabad. How many females are more than males?

Solution: This problem involves subtraction.

Number of females		$\overset{1}{\cancel{2}}$	$\overset{13}{\cancel{3}}$	$\overset{7}{\cancel{8}}$	$\overset{10}{\cancel{0}}$	4	6	3
Number of males	−	1	5	1	1	0	2	1
Difference		**0**	**8**	**6**	**9**	**4**	**4**	**2**

Hence the difference is **869,442**.

Example 3. Faraz earns Rs 16,540 in a month. How much money will he earn in 2 years?

Solution: 2 years = 24 months

			2	2	1		
Earning in a month			1	6	5	4	0
Number of months					x	2	4
			6	6	1	6	0
	+	3	3	0	8	0	0
Total Amount		**3**	**9**	**6,**	**9**	**6**	**0**

Hence Faraz will earn **Rs 396,960**

Example 4. How many boxes will be required to pack 235704 oranges, if one box contains 56 oranges?

Solution:

Number of oranges = **235,704**

Number of oranges in a box = **56**

$$
\begin{array}{r|l}
 & 4209 \\
\hline
56 & 235704 \\
 & -224 \\
\hline
 & 117 \\
 & -112 \\
\hline
 & 504 \\
 & -504 \\
\hline
 & 0
\end{array}
$$

Hence **4,209** boxes will be required.

EXERCISE 1.6

(1) Fouzia had **145,320** rupees. Her father gave her **54,304** rupees more. How many rupees Fouzia had altogether?

(2) Sobia gave a **5,000** rupee note to the shopkeeper for purchasing an electronic doll worth **Rs 2,300**. Find the remaining amount received by Sobia.

(3) A school collected funds for flood relief. **225** students of a school contributed equally **Rs 23,650**. Find the contribution by each student.

(4) A factory produced **235,806** floor marble tiles in a day . How many marbles will be produced in **32** days?

(5) Government spent **Rs 5,380,100** on gas supply in one town. What is the total amount spent on gas supply of **25** such towns?

(6) A container holds **9,475** litres of milk. How many litres of milk will **354** such containers hold?

(7) Find the number of roll required to pack **531,675** metre cloths. One roll contains 45 metres.

(8) A construction company wanted to purchase a land for housing scheme of **Rs 52,890,500**. It had **Rs 50,456,128** and borrowed the rest from the Bank. How much money did the company borrow from the Bank?

1.4 ORDER OF OPERATIONS BODMAS RULE

1.4.1 Recognize BODMAS rule, using only parentheses ( ) and carry out combined operations using BODMAS rule.

The rule of BODMAS is formed to help us in remembering the order of sloving mathematical operations. The following order must be followed by solving any mathematical problem.

B	**Brackets**	Solve within brackets ( )	
O	**of**	To multiply	
D	**Divide**	÷	
M	**Multiply**	x	
A	**Add**	+	**Add and Subtract in order from left to right**
S	**Subtract**	–	

Teacher's Note Teacher should show students that answer will be wrong if BODMAS rule is not followed

Round Brackets ( ) are also known as parentheses.
Let us understand BODMAS rule with the following examples.

Example 1. Solve by using BODMAS rule 135 ÷ 15 + 6 – 5 x 2

Solution:

135 ÷ 15 + 6 – 5 x 2 (By applying DMAS rule we first perform **Division**). (135 ÷ 15 = 9)

= 9 + 6 – 5 x 2 Then perform **multiplication.** (5 x 2 = 10)
= 9 + 6 – 5 X 10 Then perform **addition** and **subtraction** in order from left to right.

Example 2. Solve by using BODMAS 64 – (6 of 2) x 3

Solution:

= 64 – (6 x 2) x 3 First perform operation inside the bracket

= 64 – 12 x 3 → Then **multiply**

= 64 – 36 → and then **Subtract**

= 28

Example 3. Solve: 82 – 32 of 2 + (8 + 20 ÷ 4)

Solution:

82 – 32 of 2 + (8 + 20 ÷ 4) (First solve inside the **bracket**. Do **division** first.)

= 82 – 32 of (8 +) (Solve **addition** to remove the **bracket**)

= 82 – 32 x 2 + (1 + 3) (Replace **of** with **multiply** and solve)

= 82 – 64 + 13 (In the last perform addition and subtraction **in order** from left to right))

= 18 + 13

= 31

EXERCISE 1.7

A. Salman solved the following question using BODMAS
(213 - 123) + 60 x 5 (64 ÷ 8)

Solution: =(213 - 123) + 60 x 5 - 64 ÷ 8
=90 + 60 x 5 - 64 ÷ 8
=90 + 60 x 5 - 8
=90 + 300 - 8
=390 - 8
=382

Can you write down the steps that Salman followed?

Step 1: Solve ______________________

Step 2: Solve ______________________

Step 3: Solve ______________________

Step 4: Solve ______________________

Step 5: Solve ______________________

B. Solve:

(1) (9 – 8) x 18

(2) (5 x 9) ÷ 15

(3) 16 ÷ 2 + 5 x 4 – 2

(4) 32 x 21 – 42 ÷ 7

(5) 50 x 5 + (15 + 23)

(6) 7 + (15 ÷ 3 + 5) x 4 – 20

(7) 4 of 195 ÷ 13 – 54

(8) (3 x 18) ÷ 3 of 2 + 105

(9) (28 ÷ 4 + 5) x 4 – 11 of 3

(10) 5 + (42 + 7 of 2 – 2) x 8

(11) 100 of (3 x 150 ÷ 10)

(12) 60 + (72 ÷ 7 of 3 + 5) x 2

C. Add brackets to these mathematical statements so that the statements become correct.

(1) 24 ÷ 4 x 3 = 18

(2) 6 x 2 + 8 ÷ 4 = 15

1.4.2 Verify distributive laws.

There are two types of distributive laws:

(i) Distributive law of multiplication over addition with respect to multiplication.

(ii) Distributive law of multiplication over subtraction with respect to multiplication.

According to distributive law, addition or subtraction of any two numbers within brackets and its multiplication by the number outside the bracket gives the same result of the multiplication of the outer number with both the numbers and addition or subtraction of the results. Let us understand the process of verification of distributive laws with the help of examples.

For example, 4 x (2 + 5) will be equal to (4 x 2) + (4 x 5)

Example 1. Verify distributive laws.

(i) 8 x (3 + 2) = (8 x 3) + (8 x 2)

(ii) (12 – 10) x 4 = (12 x 4) – (10 x 4)

Solution: (i) 8 x (3 + 2) = (8 x 3) + (8 x 2)

Left Hand Side (LHS)	Right Hand Side(RHS)
= 8 x (3 + 2)	= (8 x 3) + (8 x 2)
= 8 x 5	= 24 + 16
= 40	= 40

LHS = RHS

Hence verified

(ii) (12 – 10) x 4 = (12 x 4) – (10 x 4)

LHS = (12 – 10) x 4	RHS = (12 x 4) – (10 x 4)
= 2 x 4	= 48 – 40
= 8	= 8

LHS = RHS

Hence verified

Example 2. Fill in the blanks.

(1) $18 \times (6 + 3) = (18 \times \boxed{}) + (18 \times \boxed{})$

(2) $(\boxed{} + \boxed{}) \times 5 = (20 \times \boxed{}) + (10 \times \boxed{})$

(3) $2 \times (5 - \boxed{}) = (2 \times \boxed{}) - (\boxed{} \times 7)$

Solution:

(1) $18 \times (6 + 3) = (18 \times \boxed{6}) + (18 \times \boxed{3})$

(2) $(\boxed{20} + \boxed{10}) \times 5 = (20 \times \boxed{5}) + (10 \times \boxed{5})$

(3) $2 \times (5 - \boxed{7}) = (2 \times \boxed{5}) - (\boxed{2} \times 7)$

EXERCISE 1.8

A. Fill in the blanks.

(1) $15 \times (5 + 3) = (15 \times \boxed{}) + (15 \times \boxed{})$

(2) $(25 - 18) \times 32 = (25 \times \boxed{}) - (18 \times \boxed{})$

(3) $(\boxed{} + \boxed{}) \times 10 = (30 \times 10) + (40 \times 10)$

(4) $(3 \boxed{} 5) \times 14 = (3 \boxed{} 14) + (5 \times 14)$

(5) $\boxed{} \times (26 + 74) = (5 \times 26) \boxed{} (5 \times 74)$

(6) $9 \times (13 - \boxed{}) = (9 \times \boxed{}) - (9 \times 5)$

(7) $10 \times (20 - \boxed{}) = (10 \times \boxed{}) - (\boxed{} \times 30)$

(8) $(\boxed{} + \boxed{}) \times 4 = (12 \times \boxed{}) + (8 \times \boxed{})$

B. Verify distributive laws.

(1) $5 \times (3 + 2) = (5 \times 3) + (5 \times 2)$

(2) $4 \times (9 - 5) = (4 \times 9) - (4 \times 5)$

(3) $(18 + 2) \times 10 = (18 \times 10) + (2 \times 10)$

(4) $10 \times (12 - 3) = (10 \times 12) - (10 \times 3)$